Relational Schema – College Course Registration System

Derived from the ER diagram for the CourseHub DBMS Level 3 project.

Notation

Bold = primary key, *italic* = foreign key. A column that is both is shown as ***bold italic***.

Relations

1. **STUDENTS**(**student_id**, student_name, email, department, semester)
2. **FACULTY**(**faculty_id**, faculty_name, email, department)
3. **COURSES**(**course_id**, course_name, credits, department, *faculty_id*)
   - *faculty_id* references FACULTY(faculty_id)
4. **COURSE_PREREQUISITES**(***course_id***, ***prerequisite_course_id***)
   - *course_id* references COURSES(course_id)
   - *prerequisite_course_id* references COURSES(course_id)
   - The primary key is the pair of both columns.
5. **ENROLLMENT**(**enrollment_id**, enrollment_date, status, *student_id*, *course_id*)
   - *student_id* references STUDENTS(student_id)
   - *course_id* references COURSES(course_id)
6. **ATTENDANCE**(**attendance_id**, attendance_date, status, *enrollment_id*)
   - *enrollment_id* references ENROLLMENT(enrollment_id)
7. **MARKS**(**mark_id**, internal_marks, external_marks, total_marks, grade, *enrollment_id*)
   - *enrollment_id* references ENROLLMENT(enrollment_id) and is also UNIQUE NOT NULL.

How each relationship was mapped

Relationship	Cardinality in the diagram	Mapping
Students **enrolls** Enrollment	1 : M	FK `student_id` in ENROLLMENT
Enrollment **for** Courses	M : 1	FK `course_id` in ENROLLMENT
Enrollment **has** Attendance	1 : M	FK `enrollment_id` in ATTENDANCE
Enrollment **has** Marks	1 : 1	FK `enrollment_id` in MARKS, with a UNIQUE constraint
Faculty **teaches** Courses	M : 1 (faculty side M, course side 1)	FK `faculty_id` in COURSES
Courses **has prerequisite** Courses	M : M (self)	Junction table COURSE_PREREQUISITES

Notes

- **Faculty–course link.** This diagram shows a course with one faculty member and a faculty member with many courses. That puts `faculty_id` inside COURSES, so there is no separate `faculty_courses` table. The 19-page report built earlier assumed an M:N link with a junction table, so it would need updating if this is the final diagram.
- **Derived attribute.** `total_marks` is derived from `internal_marks + external_marks`. It can be kept as a stored column maintained by a trigger, as the earlier report did, or computed in a view.
- **Underlined date attributes.** `enrollment_date` and `attendance_date` are underlined in the diagram, but only the `_id` attributes were treated as keys here. If the dates are meant to be part of a key, the schema needs adjusting.
- **Missing from the diagram.** There is no entity for the login accounts table (`app_users`), so it is not part of this schema.